

RESEARCH INTERESTS

- Geometrically-Consistent, Video-Based World Models
- 2D/3D/4D Computer Vision, LiDAR Moving Object Segmentation (MOS), SLAM
- Far-Field Robustness, Cross-Sensor Generalization, Architectural Lightweighting of Neural Networks

EXPERIENCE

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|---------------------------------------|----------------------|
| • ETRI Visual Intelligence Lab | Daejeon, South Korea |
| Post-Graduate Researcher | Aug. 2026 – Present |

EDUCATION

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|---|-----------------------|
| • Gyeongsang National University, Graduate School | Jinju, South Korea |
| M.S. in Computer Science and Engineering (GPA: 4.50/4.50) | Mar. 2025 – Aug. 2026 |
| • Gyeongsang National University | Jinju, South Korea |
| B.S. in Computer Science (GPA: 4.32/4.50) | Mar. 2020 – Feb. 2025 |

LANGUAGES

- **Korean:** Native
- **English:** Conversational / Capable of Q&A at international conferences / Speech contest award

PROJECTS

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| • LoRa-MOS: Density-Decay-Robust LiDAR Moving Object Segmentation (IEEE RA-L, under final review) | Feb. 2026 – Present |
| • X-MOS: Cross-LiDAR Generalization Framework (ICRA 2026) | Aug. 2025 – Jan. 2026 |
| • SwiftMOS: Real-Time LiDAR Moving Object Segmentation (IEEE RA-L) | Jan. 2025 – Jan. 2026 |
| • J.A.R.V.I.S: LiDAR-Scan-based Digital Twin & Gesture Interaction | Sep. 2024 – Dec. 2024 |
| • Finger-Pointing-based Small Robot Control (SIGGRAPH Asia 2024) | Mar. 2024 – Dec. 2024 |
| • Shrunk Reality: SLAM-based Real-Time Real-World Context Augmentation (SIGGRAPH Asia) | Jan. 2024 – Dec. 2024 |
| • SLAM-based Illegal Parking Detection System (SIGGRAPH Asia 2024) | Aug. 2023 – Dec. 2024 |
| • Low-Cost Augmented Reality Interaction (ETRI) | Jan. 2023 – Dec. 2023 |

HONORS & AWARDS

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| • Manufacturing AI Competition – Gyeongsang National University (Excellence Award) | May 2026 |
| • Young Pioneer Researcher Award – GNU Graduate School (Excellence Award) | May 2026 |
| • Glocal English Speech Contest – GNU Graduate School (Excellence Award) | May 2025 |
| • Gyeongsangnam-do Governor's Award – IT Engineering College Commencement (Top of College) | Feb. 2025 |
| • Capstone Design Competition – Gyeongsang National University (Grand Prize) | Dec. 2024 |
| • Naver 1784 School Revolution – Elementary AI Education, National 1st Place (Grand Prize) | Nov. 2023 |
| • Likelion Hackathon – Web Development, National 3rd Place (Bronze) | Aug. 2023 |

PUBLICATIONS

First author: 9, Co-author: 9

[Under Review / Revision]

- **Lee, M.**, et al. LoRa-MOS: Density-Decay-Robust LiDAR Moving Object Segmentation via Logarithmic Spatial Representation and Range-Balanced Learning. *IEEE Robotics and Automation Letters (RA-L)*, under final review, 2026.

[International Journals]

- **Lee, M.**, Kim, U., Kim, G. W., & Lee, S. (2026). SwiftMOS: A Fast and Lightweight Moving Object Segmentation via Feature Flowing Direct View Transformation. *IEEE Robotics and Automation Letters*.
- **Lee, M.**, Lee, S., & Seo, H. (2026). Enhancing Early Alzheimer's Disease Detection via Transfer Learning: From Big Structural MRI Datasets to Ethnically Distinct Small Cohorts. *Applied Sciences*, 16(2), 1004.
- Yeon, S., **Lee, M.**, Bae, J., & Lee, S. (2026). Resolving Ambiguity in Pointing Gestures Using Contextual Reasoning from Large Language Models. *Computer Modeling in Engineering & Sciences*.

[International Conferences]

- **Lee, M.**, Ha, I., Choi, S. M., Kim, G. W., & Lee, S. X-MOS: A Heterogeneous Cross-LiDAR Generalization Framework for Moving Object Segmentation. *IEEE International Conference on Robotics and Automation (ICRA) 2026*.
- **Lee, M.**, Choi, S. M., Kim, G. W., & Lee, S. HBRB-BoW: A Retrained Bag-of-Words Vocabulary for ORB-SLAM via Hierarchical BRB-KMeans. *IEEE International Conference on Robotics and Automation (ICRA) 2026* (Late-Breaking Results).
- Kang, W., Kim, M., **Lee, M.**, Kim, U., Bae, J., Lee, S., Kim, G. W., & Choi, S. M. Hierarchical Interests Modeling on Multiple Time Scales for Sequential Recommendation. *Artificial Intelligence with Biased or Scarce Data Workshop at AAAI 2026*.
- Kim, U., **Lee, M.**, Bae, J., Kim, M., Choi, S. M., & Lee, S. Privacy-Preserving Argumentative Explanations. *AAAI 2026*.
- **Lee, M.**, Bae, J., Kim, U., Choi, S. M., & Lee, S. Shrunk Reality: Augmenting Real-World Contexts in Real-Time on Realistic Miniature Dioramas. *SIGGRAPH Asia 2024*.
- Bae, J., **Lee, M.**, Kim, U., & Lee, S. SLAM-Based Illegal Parking Detection System. *SIGGRAPH Asia 2024*.
- **Lee, M.**, Bae, J., Choi, S. M., & Lee, S. Finger-Pointing Interface for Human Gesture Recognition Based on Real-Time Geometric Comprehension. *SIGGRAPH Asia 2024*.

[Domestic Journals]

- **Lee, M.**, Bae, J., & Lee, S. (2025). Watch My Wrist: Wrist Muscle Movement-based Hand Gesture Recognition for Out-of-Sight in Augmented Reality. *The Journal of the Korean Institute of Information Technology*, 23(7), 167–170. (in Korean)
- **Lee, M.**, Bae, J., Choi, S. M., & Lee, S. (2024). Plane Detection and Projection-based Filter for Enhanced Moving Object Segmentation in 3D LiDAR Environments. *The Journal of the Korean Institute of Information Technology*, 22(12), 19–27. (in Korean)

[Domestic Conferences]

- Ha, I., **Lee, M.**, & Lee, S. Image-Processing-based Hand-Tremor Compensation Pointing System for Digital Presentation Environments. *KIIT Fall Conference 2025*. (in Korean)
- Yeon, S., **Lee, M.**, & Lee, S. Low-Cost Mixed Reality (MR) System based on 3D Hand Tracking. *KIIT Fall Conference 2024*. (in Korean)
- Kim, U., **Lee, M.**, & Lee, S. Multiplayer System Combining First-Person Mixed Reality and Real-Time Strategy. *KIIT Fall Conference 2024*. (in Korean)
- **Lee, M.**, Bae, J., & Lee, S. Plane-Detection-Filter-based Moving Object Segmentation in 3D LiDAR Environments. *KIIT Fall Conference 2024*. (in Korean)
- Kim, U., Bae, J., **Lee, M.**, & Lee, S. Explainable Facial Beauty Prediction Model. *KIIT Fall Conference 2023*. (in Korean)
- Bae, J., **Lee, M.**, Kim, U., & Lee, S. SLAM-based Illegal Parking Detection System. *KIIT Fall Conference 2023*. (in Korean)